

Understanding Human Confidence in Collaborative Card Games

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Abstract

In this study we use a modified version of the collaborative card game *The Mind* to model human confidence in the gameplay of both themselves and their teammates. We assess the capability of four predictive models for perceived probability in the game, including those that take into account the number of cards remaining in play and those which involve Bayesian priors. We argue how the model variations capture significant means of human cognition. We project human confidence ratings from the best-performing models to determine the average confidence ratings corresponding the likelihood of playing a card. Our results suggest that humans rely on simplified and intuitive models to predict confidence, despite potentially overestimating likelihoods. Furthermore, the confidence rating scale of 1 to 7 is often utilized asymmetrically and varies significantly between participants.

Keywords: games; card games; confidence; priors; prediction; deduction; model complexity

Introduction

The Mind is a multiplayer, collaborative game based on intuition and confidence. The team seeks to place as many cards in their hands into the center of a growing pile, placed in strictly increasing order, for which a penalty is received for each card skipped. *The Mind* game stipulates the absence of verbal and nonverbal signals to teammates, so that the common objective is achieved as a function of each player's decisions alone and their trust in the decision-making abilities of others.

In this study, participants are asked to envision themselves in various scenarios of a modified version of *The Mind*. Through the questions we asked to the participants, we tested four cognition heuristics in the form of mathematical models most representative of a player's confidence in gameplay. Using the cognition models to obtain predictive probabilities, we construct a mapping from these predicted probabilities to confidence scores on a scale of 1 to 7, and observe the positions and variations of these confidence projections over all participants.

Our results suggest that humans favor simplistic, intuitive models over complex mathematical ones when assessing confidence for playing a card. Under such assumptions, players also seem to underutilize the full confidence scale of 1 to 7 provided, associating a wider range of probabilities to lower ratings and seemingly choosing arbitrarily above a certain perceived probability that playing their card is the optimal next move. The high probabilities from our best models

associated with participants' confidence ratings also suggest general overconfidence in gameplay in the midst of varying confidence scales for each participant.

For future study, we suggest bypassing our confidence score-to-probability conversion with different participant measurements such as asking directly for probabilities. Modifying existing scenarios to measure confidence both before and after a teammate's action could also validate our model choices in a dynamic gameplay regime.

Related Work

Gaming takes the form of entertainment, sport, teaching, training, and research, as enumerated in Schwabe (1994). They involve a setting of restricted knowledge where players attempt to achieve their objectives. Games offer investigators an environment to understand policy and decision-making. They are nevertheless underutilized, being dismissed as not scientifically rigorous (Schwabe, 1994). Despite being difficult to design, good games offer increased understanding of assumptions and the discovery of hidden ones (Schwabe, 1994), thus contributing to cognitive science research.

Given the unknowns of a game, there arises a proposal for decision-making: predicting a probability of success, which can in turn inform a decision. One decision to be made is if a prior is to be considered. The argument against using a prior in modeling is evidenced by a prevalent human cognitive bias for which decisions are made without sensitivity to the prior probability of outcomes (Tversky & Kahneman, 1974). Instead of relying on priors, there is evidence for an "illusion of validity" among humans, where the prominence of observing a particularly representative example of an event can dominate a prediction, eclipsing other factors such as priors at the cost of accuracy in predicted probability (Tversky & Kahneman, 1974).

There is support for using Bayesian models in the context of a game, however. When tracking a large number of future possibilities (or in the our game, the resolution of the hidden cards of the teammates), humans prune sections of the decision tree to alleviate computational burden. This pruning process is liberal in practice; in fact the process is consistently Pavlovian on negative stimuli (i.e. disposing of large branches given one experience of negative feedback in the branch) (Huys et al., 2012). This insight suggests prior disposal of possibilities which are irrevocably negative or impos-

sible, which exist in Bayesian models of sufficiently restricted games.

The limiting focus of this study is collaborative games. Collaborative games reach greater in breadth in multiplayer online games, but they are better analyzed in the simple framework of board games (Zagal, Rick, & Hsi, 2006), which we do in this study. Within this framework for a successful collaborative game, it should be feedback-oriented and feature individual action within its collective objective, so players deal with the complexity of balancing individual utility versus team utility (Zagal et al., 2006). Although these considerations are largely met in the modified version of *The Mind* game described in the following section, the analysis also notes qualities of the game that enhance its playability: the players should care about the satisfaction of a collective result, and potentially, this requires designing multiple roles and responsibilities, which makes successive participation in the game enjoyable (Zagal et al., 2006).

There are a number of considerations necessary to model confidence ratings from perceived probabilities, which this study collects on a scale (1-7) related to, but not explicitly corresponding to, the probability of success. It is often inadequate to consider perceived probability as the only piece of this model (Navajas, Hindocha, Foda, et al., 2017). In fact, for most probabilities assessed by a human, the true probability is lower, especially for probabilities above the 0.5 threshold, demonstrating evidence of overconfidence in humans (Lichtenstein & Fischhoff, 1977). This discrepancy between the perceived probability and true probability is greater when the task at hand is more difficult, and naturally also greater for untrained subjects than trained subjects (Lichtenstein & Fischhoff, 1977). One factor in predicting confidence besides perceived probability is suggested to be perceived uncertainty in a decision (Navajas et al., 2017), which offers a counterpoint to a plain assumption that people overestimate. Observing perceived probability and perceived uncertainty contribute to individual differences in confidence ratings.

Confidence rating is also affected by the post-decision stimulus of an action, prediction, or decision. In games, this exists if a play results in a mistake or is evidenced as a mistake by players in their reactions (Moran, Teodorescu, & Usher, 2015), though the latter factor is not present in static scenarios. The study we conduct consists not only of predicting confidence ratings for a potential choice of playing or not playing with no other stimulus, being Individual Evaluation scenarios described in the following section, but also of scenarios of predicting confidence ratings from the stimulus of revealing another person's played card, constituting the Team Evaluation scenarios.

Game Setup and Data Collection

Gameplay

For the modified version of *The Mind* that we asked our participants to play, we developed the following simplified set of rules.

1. The game is played with 50 cards, numbered 1 to 50.
2. Each of the three players is initially dealt three cards.
3. Initially, the center card pile is empty.
4. Players, without communicating verbally, must place their cards on the center pile such that only higher cards can be placed on the existing center card.
5. At any point, if a card is played by a teammate that is larger than any of your cards, you must discard those cards which can no longer be played.
6. The objective of the game is to maximize the number of cards played onto the center.

Data

We gathered 9 participants and had them play a couple of actual games with one another, clarifying any confusions along the way. Then, we provided them with two types of gameplay scenarios, asking for a confidence estimate between 1 (lowest) and 7 (highest) for each. The general scenario types and their abbreviations within this paper are as follows:

- **Individual Evaluation. (IE)** Suppose the cards you started with are a, b, c . Your teammates have x and y cards left. The current top card in the center pile is d . How confident are you (1-7) that your lowest playable card is the lowest playable card amongst all three of you?
- **Teammate Evaluation. (TE)** Suppose the cards you started with are a, b, c . Your teammates have x and y cards left. The current top card in the center pile is d . The first/second teammate plays a card e . How confident are you (1-7) that e was the lowest playable card amongst all three of you?

Some IE scenarios were structured so that a real game was being simulated between three different players, and different participants were provided with different players' perspectives. For example, three players would all be provided with a scenario where $d = 26$, but x and y would vary based on the other teammates' number of cards above d . We refer to these scenarios as *IE Group* scenarios.

In the TE scenarios, we emphasized that while participants could assume that each teammate would play their own lowest playable card, that it was up to the participants themselves to determine how confident they were in that teammate's play, without knowing the third player's cards.

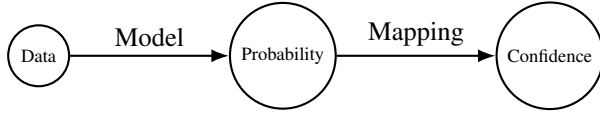
From each of the 9 participants, we collected 9 IE scenario confidence ratings, 3 of which were IE Group scenarios. We also collected 8 TE scenario confidence ratings.

Confidence Prediction

We first focus on the task of predicting a participant's confidence rating between 1 and 7 in a given scenario. All models first compute a probability that is intended to model the likelihood that the player would actually play the given card in the

scenario before any teammates attempt to play their cards. In subsequent sections, we discuss our results and methods for translating these probabilities to the confidence scores that were asked of the participants.

In our study, we only worked with models that predict a probability value, which can then be translated to or from a confidence rating between 1 and 7. This enforces a prediction architecture shown in Figure 1, rather than directly predicting a confidence rating between 1 and 7 from the data.



Models

- Division
- Naive
- U Bayes
- W Bayes

Mappings

- Linear
- Inverse Sigmoid

Figure 1: Confidence Prediction Procedure

While this method precludes models that can directly produce a confidence rating, thus bypassing the probability prediction step, this heuristic assists our later methods of mapping confidence ratings to likelihoods of participants playing their lowest card in a given scenario.

Modeling Cognitive Hypotheses

In studying our different models for predicting participants' probability assessments in gameplay, we assess the viability of hypotheses for how humans think with probability in decision-making. To this end, we hypothesized four different models, each intended to capture a subset of considerations and keys to thinking a participant might apply when playing our game.

- Division: The most simple (baseline) model. It predicts a probability from 0 to 1, increasing linearly as the participant's lowest (playable) card approaches the center card (i.e. decreases and moves away from 50).

Keys: *Reductive, Intuitive, Trend*

- Naive: This model assumes independence between each remaining card held by each player. All cards that have yet to be played (but that can still be played) have an independent and identically distributed placement among the cards greater than the center card.

Keys: *Analytical, Representative, Estimation*

- U (Unweighted) Bayes: This model assumes every card is equally likely to be the minimum card for each player. However, once a scenario is introduced, it marginalizes

over the minimum card choices that could still be valid for each teammate in the range of remaining cards (i.e. disposes of invalid combinations) before computing probability.

Keys: *Cumulative, Deductive, Priors*

- W (Weighted) Bayes: This model introduces weighted priors for the minimum cards into predictions (e.g. a 1 is much more likely to be the minimum card in a starting hand than a 42). The marginalization procedure is the same as in U Bayes.

Keys: *Salient, Human Priors*

The Division and Naive models for thinking about the game scenarios follow observations that humans make decisions without factoring in priors (Tversky & Kahneman, 1974). An outcome which is considered representative can prove solely influential in predicting confidence (Tversky & Kahneman, 1974). This testifies against the Naive model, where distributions of remaining cards which are perceived to be IID dominate judgment, instead of a mathematical analysis based in expected values. The same observation manifests in the focus on saliency in the the weighted Bayesian model.

The process of marginalization of some form as a response to the actions of a game may be found in the observations of pruning (Huys et al., 2012), when a large possibility space is restricted upon new stimuli. The two Bayesian models use marginalization (elimination of possibilities) as a response to the stimulus of the center card (the last card being played), a martingale-like simplification to the presence of the most recent stimulus.

Parameters of Interest We limit ourselves to the following notation which more-or-less captures the relevant parameters of the game which influence the confidence assessment: As introduced in the previous section, we let a, b, c be the three cards (in increasing order) that the participant is initially provided with. To simplify notation, we let L be the lowest card (among a, b, c) that the participant can legally play. We also let x and y be the number of remaining cards the two teammates have (in no particular order), and d be the current center pile's top card. Furthermore, in the TE scenarios, we define e to be the card played by the teammate.

Division (Baseline) Model

In the IE scenario, this model predicts a probability of

$$\frac{50 - L}{50 - (d + 1)},$$

while in the TE scenario, this model predicts a probability of

$$\frac{50 - e}{50 - (d + 1)}.$$

Both predictions are 1 exactly when the card $d + 1$ is played, and 0 if the card 50 is played.

Naive Model

This model factors into account the number of cards each teammate has. Very similar to the Division Model, the probability that any one card is between L and 50 given that it is uniformly distributed at random between d and 50 is $\frac{50-L}{50-d}$. For IE, since the opponents share $x+y$ cards, the model outputs a probability of

$$\left(\frac{50-L}{50-d}\right)^{x+y}.$$

When considering the TE Scenario, we can assume without loss of generality that the first teammate played. We know that this teammate played their lowest card, so barring the case where they play a card e lower than L (in which case the model immediately predicts 0), the only cards we need to estimate are that of the second teammate. We assume that this teammate has $\frac{x+y}{2}$ remaining cards in their hand, thereby predicting

$$\left(\frac{50-e}{50-d}\right)^{\frac{x+y}{2}}.$$

Bayesian Model

Our final model initializes three $50 \times 50 \times 50$ tables of weights, one to represent the perspective of each player. Although our study only focused on predicting probabilities after at most a single action is taken in our scenarios, this would allow us to predict probabilities from the perspective of each player when multiple-play scenarios are introduced.

Let the players be numbered 0, 1, 2. For a given player numbered P , let B represent their corresponding table. The entry $B[i, j, k]$ represents the probability that player P 's lowest card is i , player $P+1 \pmod{3}$'s card is j , and player $P+2 \pmod{3}$'s card is k . Since we will be summing over multiple indices in our table, we adopt the notation

$$B[i_1 : i_2, j_1 : j_2, k_1 : k_2] := \sum_{i=i_1}^{i_2-1} \sum_{j=j_1}^{j_2-1} \sum_{k=k_1}^{k_2-1} B[i, j, k].$$

If any of these summations involve just one index (e.g. $i_1 = i_2 - 1$, we omit the second (end) index.

Parameter Initializations We considered two different initializations of this model, generating two different prediction probabilities. In the *unweighted Bayesian model* (U Bayes), we initialize all $B[i, j, k]$ to 1, so that every card has an equal probability of being the lowest card for a player, and the players' cards are independent of their teammates'.

In the *weighted Bayesian model* (W Bayes), we assume that each player's three cards are independently and identically distributed uniformly at random from 1 to 50, so that the probability ratio of i_1 being the lowest card versus i_2 being the lowest card is approximated by $\frac{(51-i_1)^3 - (51-(i_1+1))^3}{(51-i_2)^3 - (51-(i_2+1))^3}$. As a result, to preserve ratios and independence between different players' cards, we initialize B such that

$$B[i, j, k] = \prod_{l \in \{i, j, k\}} ((51-l)^3 - (51-(l+1))^3).$$

Model Predictions For the IE Scenario, we divide the number of still-valid scenarios where both teammates' lowest cards are above L by the number of still-valid scenarios with no restrictions. Thus, the predicted probability is

$$\frac{B[L, L+1 : 51, L+1 : 51]}{B[L, d+1 : 51, d+1 : 51]}.$$

For the TE Scenario, there is one degree of unknown, that being the card held by the second teammate (who didn't play the card e). The second teammate could hold any card from $d+1$ to 50, inclusive except for e . Thus, the predicted probability is

$$\frac{B[L, e, e+1 : 51]}{B[L, e, d+1 : e] + B[L, e, e+1 : 51]}.$$

Fitting Model Predictive Probabilities to Confidence

Once our model provides probability estimates, we use either a linear or inverse sigmoid mapping to reconstruct a confidence rating. To fit such parameters, we first averaged all 9 participant confidence estimates for each scenario. In the IE Group scenarios, however, because we had assigned different questions to different participants, we only had 3 responses to average. Next, we regressed the model prediction *probabilities* on the average confidence ratings for each scenario, fitting the following parameters for each mapping:

- Linear: $\text{prob} \sim a \cdot \text{confidence} + b$
- Inverse Sigmoid: $\text{prob} \sim \frac{a}{1 + \exp(-c \cdot \text{confidence} + d)} + b$. Because the mapping from the data to probability to confidence would require the inverse of this function, we refer to the mapping as an *inverse sigmoid*.

For the purpose of this study, we also limit the possible models for predicting probabilities from data to the ones described in this section. However, each one not only is based off of reasonable assumptions of human cognition in gameplay, but also produces a significantly different prediction from the other models. A summary of the features of each model is provided in Table 1.

Model	Division	Naive	U Bayes	W Bayes
# Cards Left	N	Y	N	N
Good Resolution	N	N	Y	Y

Table 1: Comparison of model considerations

The number of cards left to be played by the two opponents is considered by the Naive model. The other models focus only on the minimum card that remains in the hand of each player.

Good resolution means the model determines each probability $B[i, j, k]$ (where B is defined in the Bayesian model description) as part of its prediction procedure. This is the Bayesian quality of the model, in its holistic consideration of the triples (i, j, k) for the minimum available card playable by the players, respectively. *Good resolution* does not necessarily hold superior. Huys (2012) demonstrates that humans consciously dispose of complexity while processing (exponentially) large sets of possibilities; a quality which Bayesian models fail to capture.

For both the linear and inverse sigmoid mappings, we minimize the residual sum of squares between predicted probabilities and average confidence ratings, with an additional regularization term of $0.0001(c^2 + d^2)$ for the inverse sigmoid.

Confidence Prediction Results

In Table 2, we see the results of regressing each model’s prediction probabilities on the average participant confidence rating in the IE scenarios. In Figure 2, we also visualize the best-fit models for both linear and sigmoid mappings using the IE scenario data. Note that there are 15 total data points, 6 from the non-IE Group scenarios, and 3 IE Group Scenarios each with 3 different perspectives, yielding 9 total IE Group data points.

Model	Linear	Inverse Sigmoid
Division	0.676	0.925
Naive	0.764	0.883
U Bayes	0.680	0.866
W Bayes	0.633	0.690

Table 2: Coefficients of determination for IE scenarios

We include the results of regression for the TE scenarios in Table 3. Because the TE scenarios provided to participants generally resulted in only confidence ratings between 1 and 4, both linear and inverse sigmoid mappings performed poorly. The graphs for the corresponding regressions are included in the appendix, but the analysis for determining the best predictive models under each mapping (linear or inverse sigmoid) that follows focuses on the IE scenarios, because of their comprehensiveness in participants’ confidence ratings between 1 and 7.

Model	Linear	Inverse Sigmoid
Division	0.422	0.476
Naive	0.250	0.369
U Bayes	0.422	0.476
W Bayes	0.222	0.342

Table 3: Coefficients of determination for TE scenarios

Analysis

The predictions for the Naive and Division models in the IE scenarios result in the highest R^2 values against the average

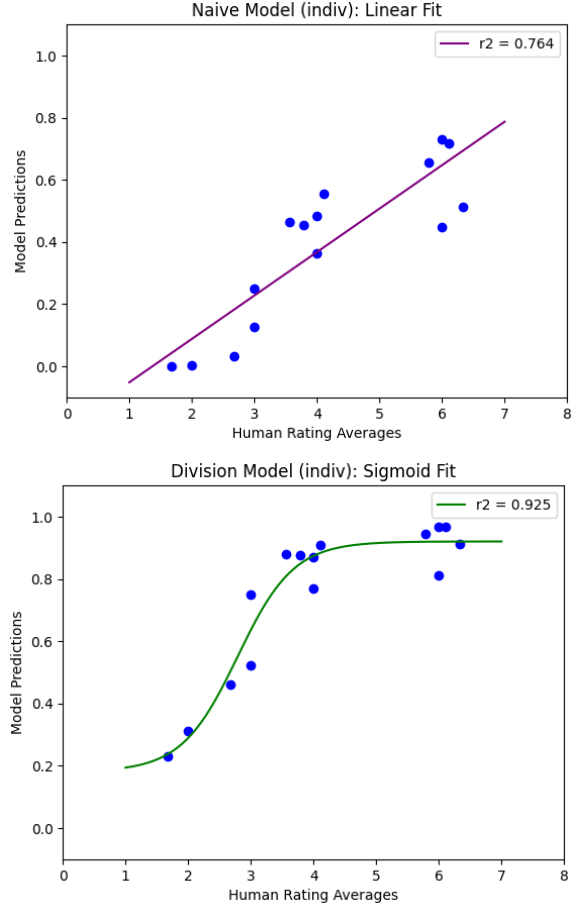


Figure 2: Mappings between Confidence and Probabilities with best R^2 for IE Scenarios

human confidence ratings, suggesting that humans sideline priors in favor of simpler non-Bayesian predictions. There is no evidence that participants compute high exponents (> 1) of probabilities or maintain complicated priors to determine the likelihood of one card being the minimum card over another.

The high R^2 for the inverse sigmoid mapping from Division model probability to confidence reveals two observations about how participants used high confidence ratings between 4 and 7. Participants do not use this upper half of ratings unless they are quite confident (probability > 0.7) in their predictions. However, once this high probability threshold is attained, participants use the higher ratings from 4 to 7 liberally despite no significant difference in predicted probability for the IE scenarios.

Compared to the other three models, the Division model consistency predicts the highest probabilities under each scenario. The fact that the Division model best represents the underlying probabilities for human confidence ratings suggests the potential presence of overestimation in human assessments of probability.

Another indication of overconfidence in the Division

model are within the three IE Group scenario results, considering the probabilities for each perspective which in reality should sum to 1 (which player's smallest card is the true smallest), but where the Division model predicts for all three IE Group scenarios probabilities from each player's perspective that collectively sum to greater than 1.

Projecting Confidence Ratings

We now turn to answering the following question: If a participant predicts a probability of p that they will play their smallest card or a predicts a probability of p that another player's play onto the pile is correct (smallest), what confidence rating r would reflect this predicted probability p ?

We also seek to answer a related question: What is the distribution of the confidence ratings r over all participants?

We answer this question for $p = 0.2, 0.5$, and 0.8 , which are our benchmarks for low confidence, medium (threshold for playing) confidence, and high confidence, respectively. We answer this question using data from both IE and TE. However, since we don't have predicted probabilities, we consider only the best models for linear and sigmoid mappings in their respective departments, the Naive model for linear and the Division model for sigmoid. We fit for *each individual* (9 total) in this study the observed confidence ratings (integers between 1 and 7, inclusive) to the model's predicted probabilities. There are 17 confidence ratings given by each participant, 9 from IE scenarios and 8 from TE scenarios. Results from all eight model-mapping combinations can be found in the appendix.

Note the presence of two types of outliers. First, in both models, confidence predictions either less than 1 or greater than 7 may arise. These are acceptable because they indicate a bias in the model of choice. A confidence rating projection of $r = -1.4$ for a probability $p = 0.2$ on a scale from 1 to 7 indicates the possibility that the participant never considered a probability as low as 0.2 in any of the scenarios. Second, in the sigmoid model, confidence predictions which are undefined exist, due to the limited domain of the inverse sigmoid function. This is equivalent to a confidence projection of $-\infty$ or ∞ where the former is below the lower end of the domain and vice versa. We provide substitute confidence projections of -1 and 9 for these two scenarios, and display them as outliers in the results.

Results of Projection

The Division model, when fitted to a sigmoid function for each of the 9 participants, estimates on average a confidence rating of -0.20 for $p = 0.2$, 2.76 for $p = 0.5$, and 4.63 for $p = 0.8$. This is described in Table 4.

p	μ_c	σ_c
0.2	-0.1976	1.2355
0.5	2.7624	0.4860
0.8	4.6282	1.7182

Table 4: (Division Model, Sigmoid) μ_c is the mean projected confidence for the 9 participants, σ_c is the standard deviation

The distribution of the confidence ratings projected is shown in Figure 3. According to the Division model, for six of the nine participants, the probability $p = 0.2$ is too low to correspond to a confidence rating. For one participant, the probability 0.8 is too high to correspond to a confidence rating. These constitute the outliers.

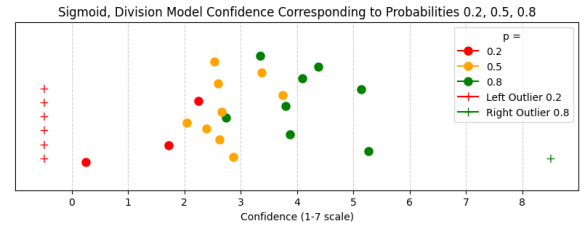


Figure 3: Projected confidence ratings for 3 instances of p for each of the 9 participants

We visualize each participant's confidence projections for the three probabilities in Figure 4. The minimum span of the confidence projections (between $p = 0.2$ and $p = 0.8$ for one participant is 2.15. The maximum span is from $-\infty$ to ∞ (indicating that both $p = 0.2$ and $p = 0.8$ are not in the domain of the inverse sigmoid).

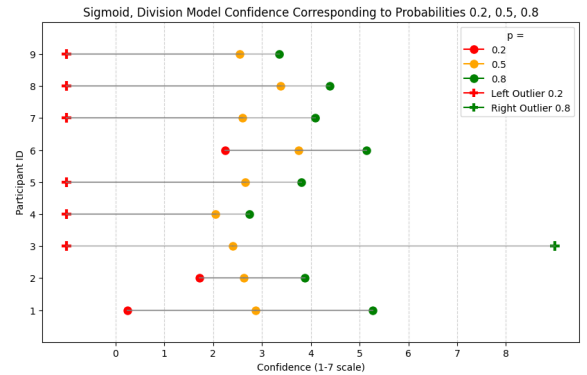


Figure 4: Projected confidence for sigmoid (logistic) fitting on Division (baseline) model, featuring 9 participants

The Naive model, when fitted to a linear function, estimates on average a confidence rating of 2.51 for $p = 0.2$, 5.12 for $p = 0.5$, and 7.72 for $p = 0.8$. This is described in Table 5.

p	μ_c	σ_c
0.2	2.5140	0.4276
0.5	5.1168	0.3179
0.8	7.7196	0.5430

Table 5: (Naive Model, Linear) μ_c is the mean projected confidence for the 9 participants, σ_c is the standard deviation

We visualize each participant’s confidence projection in Figure 5. According to the Naive model, there are no outliers, which is consistent with a fit to a linear function. The span of the confidence projections (which is proportional to the coefficient a , the slope) has over the participants a minimum of 4.44 and a maximum of 6.76.

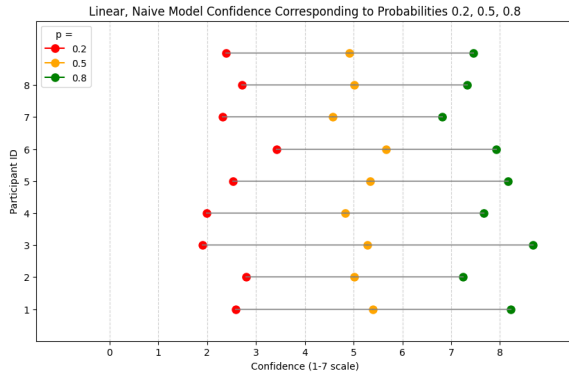


Figure 5: Projected confidence for linear fitting on Naive model, featuring 9 participants

Analysis

The results of the projected confidence for the simple, intuition-based Division model in Figure 4, where participants reserve higher confidence ratings (5 or more) for only the highest ($p > 0.8$) probabilities, suggest that humans may reserve a large amount of the confidence scale to represent high probability (over-representation of the perceived probabilities 0.95, 0.99, 0.995, 0.997, ...).

The results of the Division model show the significance of uncertainty in confidence prediction (Navajas et al., 2017) in full, where the variance of $p = 0.2$ and $p = 0.8$ confidence projections are high (see Table 4) and are even higher than stated (due to outlier normalization), evidenced by some $p = 0.5$ projections surpassing other participants’ $p = 0.8$ projections and some $p = 0.2$ projections surpassing other participants’ $p = 0.5$ projections.

The large difference between the narrowest span and the widest span is prominent evidence for the different ways people embed their perceived uncertainty (Navajas et al., 2017), but it may be overstated by the inverse sigmoid fit.

The Division model potentially misses the point (by being too overconfident) where for low probabilities like $p = 0.2$, the majority of projections are outliers, indicating that perceived probability $p = 0.2$ is off the table for these partici-

pants. This is despite the existence of clear examples of true $p \leq 0.2$ in the tested scenarios.

The Naive model under linear fit in Figure 5 rarely produces probabilities higher than 0.8 for any scenario tested, which is reflected by nearly every participant’s $p = 0.8$ confidence rating being off the right side of the standard 1 to 7 scale. The nature of the linear fit is reflected in the confidence for $p = 0.5$ as the mean of the projections of $p = 0.2$ and $p = 0.8$, which also yields more consistent $p = 0.2, p = 0.8$ confidence spans, ones which appear more reasonable than the sigmoid fit counterparts.

If the Naive model was the most representative model for predicting human confidence rating, then the mean of its $p = 0.2$ projections, 2.51 (Table 5) suggests in contrast to the Division model, that humans dedicate a significant part of their confidence rating scales to lower probabilities, and more so than to higher probabilities.

Limitations

In Gameplay and Setup

Setup and data collection details exhibit limitations in collaborative game design, where some concepts are captured in our designed scenarios, but others are ignored due to complexities for models and the learning curve for participants.

Gameplay is simulated as disconnected instances of the game in each scenario, rather than a sequence of game situations, one following another. This makes the heuristic for participants potentially closer to analytical of the scenarios than an iterative feedback-response approach that can be accounted for in a Bayesian model.

In an analysis of a collaborative game, two lessons stand out: players tracing payoffs to their decisions in a clear way and the ability for players to make actions without the consent of the team (while still working for team welfare) (Zagal et al., 2006). The former is enumerated in the rules explained to participants: knowing when a card played is wrong (players being required to dispose their cards), and thus, also being able to deduce the correctness of a played card. The latter lesson intends to capture how the collaborative objectives of a game must be achieved without guardrails against a player getting things wrong. This is an important factor into why the game scenarios designed facilitate predictions of confidence.

In the same paper, a luxury component of collaborative game design is suggested: where players differ in game capacity and responsibility (Zagal et al., 2006). While this suggestion is valid in enhancing the benefits of the actions of each player for their teammates, a study conducted on the original version of *The Mind* would introduce too many parameters. In addition, the complexity of a multitude of roles is difficult to teach to participants, especially if they use this knowledge primarily to assess disparate scenarios.

Any studies conducted with large numbers of parameters are best observed in the summative property of gameplay, where intuition and experience allows a player to distill the complexity of the game in decision making.

Teammate Evaluation Scenarios

The practical limitation of using the TE scenarios lies in their skewing towards low probabilities that were reflected in low confidence ratings for these scenarios. For example, with five cards left to be played and the center card at 30, the probability that the card a participant holds, say 40, is smaller than any card remaining, is low. Low diversity of such TE scenarios leaves them largely inadequate ($R^2 < 0.5$) for predicting the best of the four cognitive-probabilistic models and the corresponding mapping. This limitation does not apply to confidence rating projections; in fact, complementing TE with IE (which is biased towards higher probability scenarios) produces a more thorough usage of the 1-7 scale.

A theoretical limitation of using TE scenarios alone for modeling is that the teammate's play does not serve its intended purpose of providing a stimulus that can change a prior in a Bayesian model (Moran et al., 2015), nor does the teammate's play constitute an influential representative example (Tversky & Kahneman, 1974) that comes to define decision making under a simpler model of cognition. Instead, the TE scenarios are static in the same way that IE scenarios are. There are simply more pieces of relevant information than the IE scenarios to process (the original center card and the card the teammate played versus the original center card only). The deviation from the intention of TE scenarios results in large part from collecting data from handpicked scenarios rather than from live gameplay.

Discussion

We observe in this study that within the context of the gameplay scenarios presented, the best fit model is the Division model. The Division model is also the simplest, coming with the implications that participants

- exhibit overconfidence in prediction probabilities, consistent with (Lichtenstein & Fischhoff, 1977).
- use ratings between 1 and 4 for scenarios in the bulk of the range of likelihoods (0.2 to 0.8)
- give highly variable and seemingly arbitrary confidence ratings between 4 and 7
- vary significantly from one another in how much of the confidence scale they use.

The third point, which suggests that confidence ratings from 4 to 7 generally refer to the range of high probability, may be beyond the mathematical truth in the game scenarios themselves. Instead, two game scenarios which are both in truth high probability could show enough distinction in wording or card numbers involved so that they elicit significantly different ratings (say 4 and 7) from the participants.

Each of three later implications provide suggestions to how confidence scales as a measurement of perceived probability may be improved. The second point and fourth points

suggest standardization of confidence ratings for each participant based on either the entire range or some intermediate range of their observations, and the third point suggests a post-collection transformation of the higher (more confident) end of the scale so it decreases in span (from [4, 7] to [4, 5]).

In the cognitive modeling aspect, Bayesian and mathematically intensive models may not have failed to assess true probabilities. It's unclear if any one of the mathematically intensive models is reasonably close to capturing the ground truth probability of a scenario. Instead, the implications of this study suggest only that a simply construed baseline model is better, or at least as good as the more mathematical models in predicting how humans would rate these probabilities.

Future Directions

The model and mapping choice was ultimately determined by a regression of predicted model probabilities onto confidence ratings. Collecting data from participants on their own predicted probabilities (in addition to their confidence ratings) in future studies could validate both choices and verify our confidence rating projection estimates. Furthermore, they could assist with standardization of confidence ratings between each participant before a linear or inverse sigmoid regression is performed.

To address the static characteristic of the TE scenarios and their inability to induce a prior update in Bayesian models (Moran et al., 2015), we suggest a related setup meriting future study. First, the scenario would posit an IE setup and measure the participant's confidence. The participant would then receive a stimulus involving a teammate playing a card, and be asked about their new confidence rating for their own card. Measuring how the rating changes could also verify whether a simplifying assumption such as the Division model could accurately handle dynamic updates to probabilities.

Code (Google Colab)

The runnable and editable (make a copy) code for this study can be found at the following Google Colab. <https://colab.research.google.com/drive/1DPJ1aeTLCeSCDg3qaYcxfvEmq9DF7Ro?usp=sharing>.

Contribution Statement

Problem formulation, data collection, and model construction were performed in equal parts between each group member. Alan focused mostly on the first question of confidence prediction, where Aaron focused on the second question of projecting confidence ratings and developing the code for this project. All other sections in the report were written and reviewed jointly.

Acknowledgments

We express great thanks to the always-encouraging Dr. Tenenbaum and the fabulous TAs who got this project on its feet: Alicia, Tara, and not the least, Ced.

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Appendix

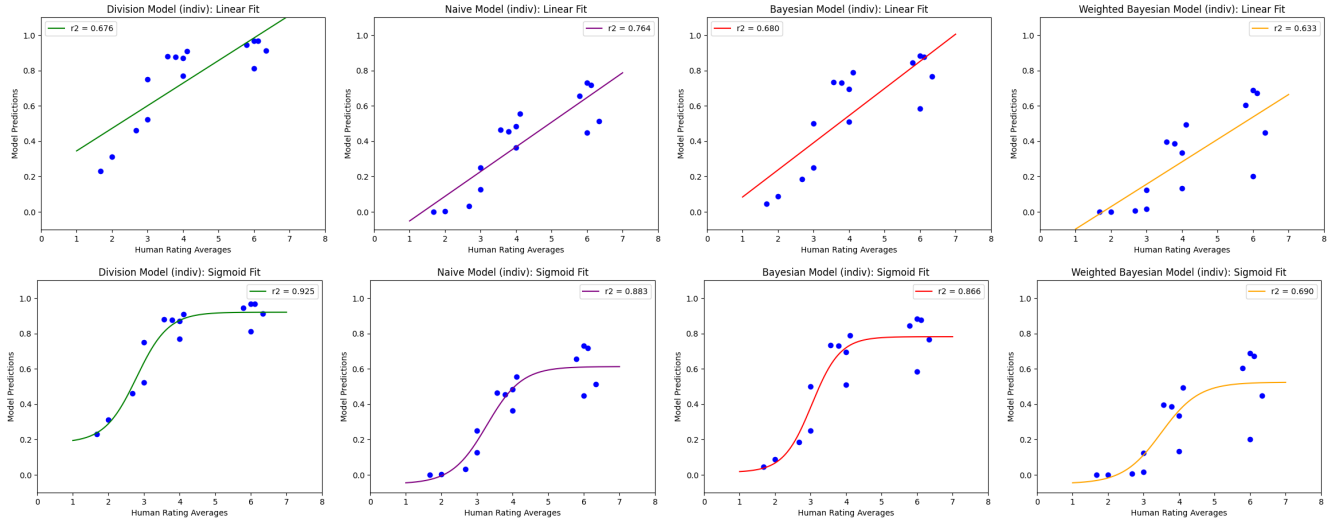


Figure 6: IE Scenario Regression Mappings

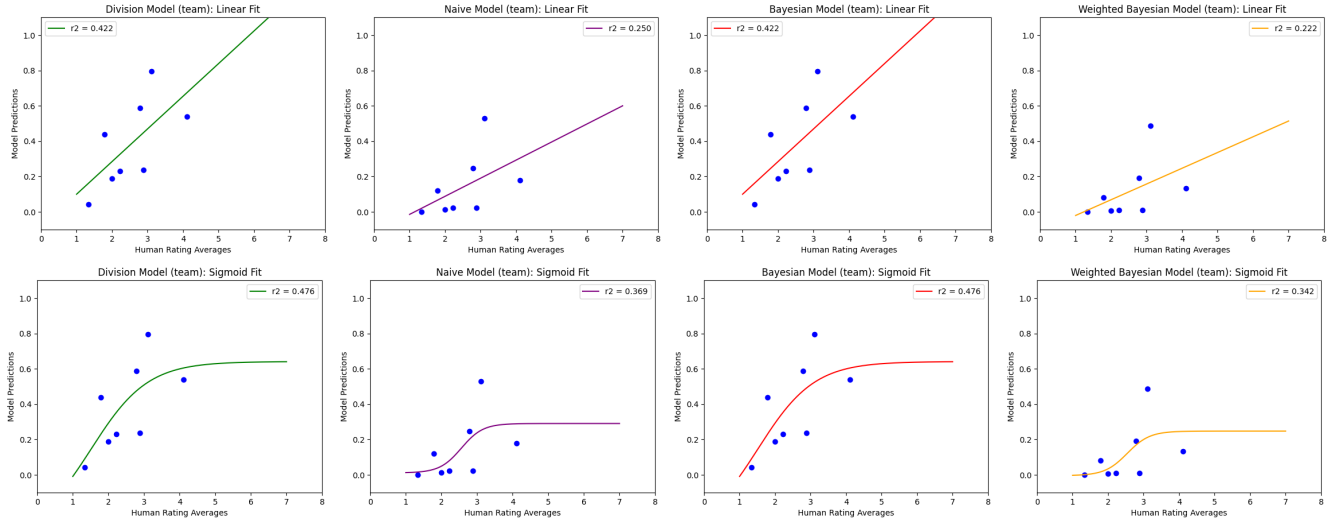


Figure 7: TE Scenario Regression Mappings

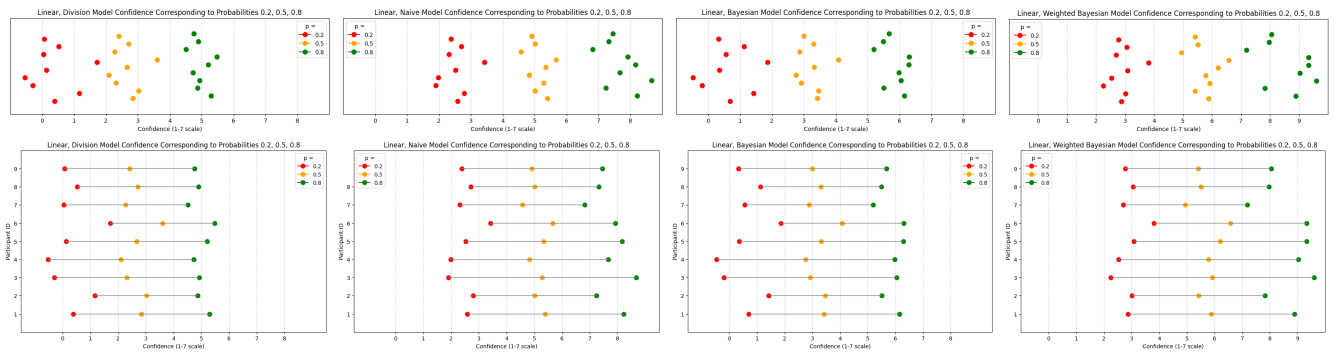


Figure 8: Linear Model Confidence-to-Probability Projections

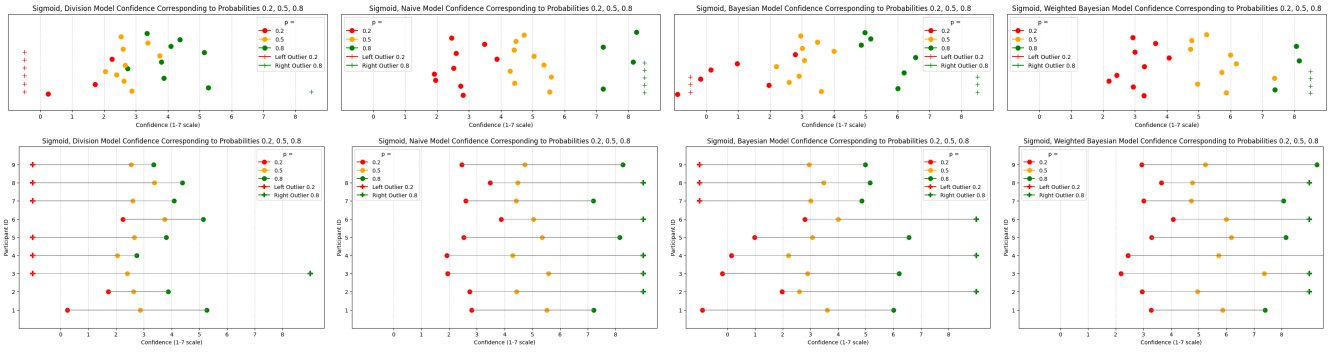


Figure 9: Sigmoid Model Confidence-to-Probability Projections